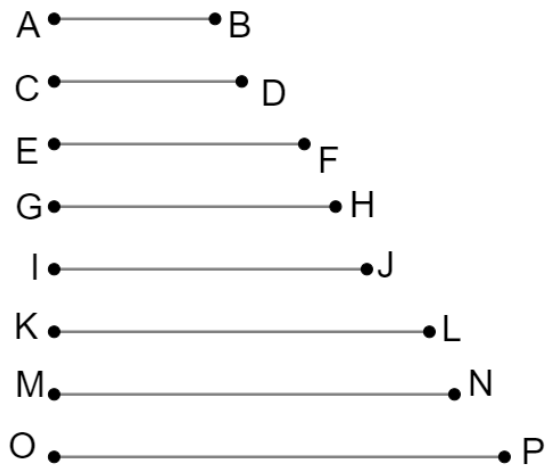
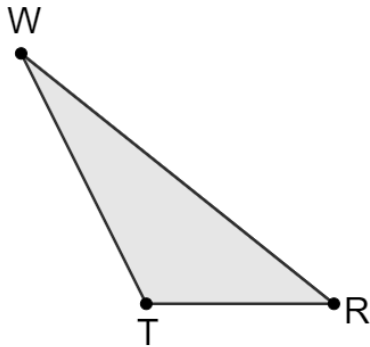


Geometry
Unit 5
Lesson 1 Practice

Name _____
Date _____
Period _____

Triangle WRT and 8 line segments are shown.



Use patty paper, a ruler, or edge of a piece of paper to determine which line segments have the same measure as each side of triangle WRT .

Complete each.

1. $WT =$ _____
2. $RT =$ _____
3. $WR =$ _____
4. Construct a triangle using the line segments that have the same measure as each side of triangle WRT . Name the new triangle XYZ .

5. Complete the statement.

The three sides of triangle WRT are congruent to the three sides of

triangle XYZ because $WT =$

- ☐ $XY,$
- ☐ $YZ,$
- ☐ $XZ,$

$RT =$

- ☐ $XY,$
- ☐ $YZ,$
- ☐ $XZ,$

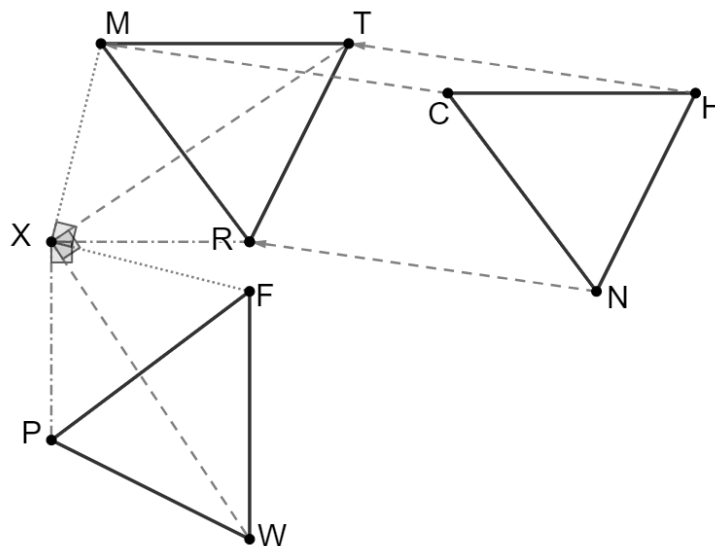
and

$WR =$

- ☐ $XY.$
- ☐ $YZ.$
- ☐ $XZ.$

6. Determine if the angles of $\triangle WRT$ and $\triangle XYZ$ are congruent. Explain how you determined your answer.

A series of transformations were applied to $\overline{CH}$, $\overline{HN}$, and $\overline{NC}$.



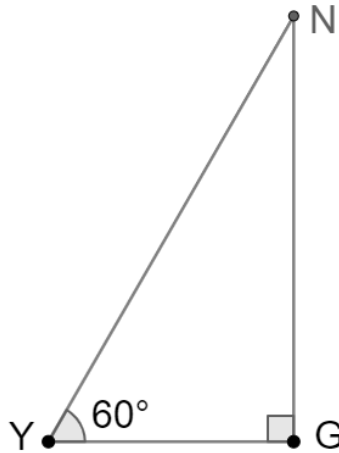
Complete the statements.

7. The first transformation applied to $\overline{CH}$, $\overline{HN}$, and $\overline{NC}$ is _____. This results in $CH =$ _____, $HN =$ _____ and $NC =$ _____.
8. Therefore, by the definition of _____, $\triangle HNC \cong \triangle TRM$.
9. The second transformation applied to $\overline{MR}$, $\overline{RT}$, and $\overline{TM}$ is _____. This results in $MR =$ _____, $RT =$ _____ and $TM =$ _____.
10. Therefore, by the definition of _____, $\triangle TRM \cong \triangle$ _____.

Geometry
Unit 5
Lesson 2 Practice

Name _____
Date _____
Period _____

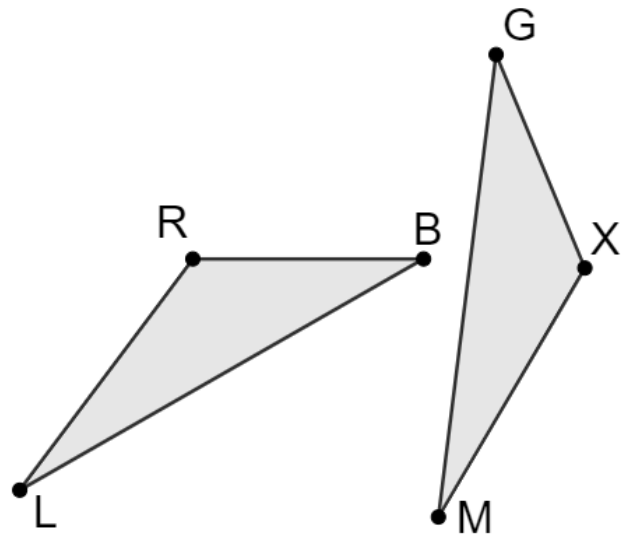
Triangle NGY is shown where $m\angle Y = 60^\circ$ and $m\angle G = 90^\circ$.



Complete the steps in the construction of $\triangle HRT$ so that $\triangle HRT \cong \triangle NGY$ using two angles of $\triangle NGY$ and their included side.

- | | |
|----|---|
| 1. | Copy _____ to construct $\overline{RT}$ so that _____ $\cong \overline{RT}$. |
| 2. | Copy _____ to construct $\angle T$ so that _____ $\cong \angle T$. |
| 3. | Copy _____ to construct $\angle R$ so that _____ $\cong \angle R$. |
| 4. | Label the intersection of the sides of the two angles H . |
5. Describe how to use geometric tools to determine if $\overline{YN} \cong \overline{TH}$, $\overline{GN} \cong \overline{RH}$, and $\angle N \cong \angle H$.
6. What postulate or theorem does the construction illustrate?

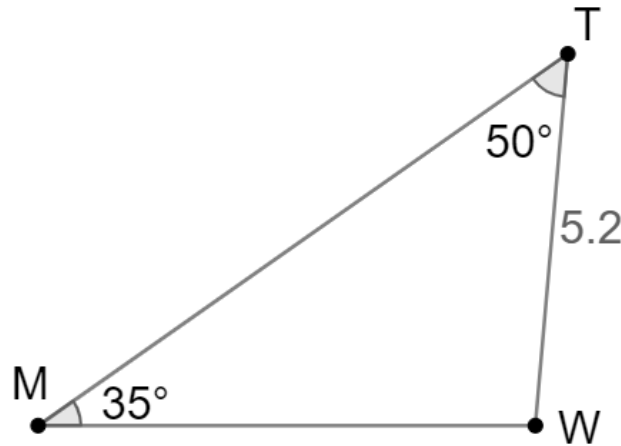
Triangles RBL and GXM are shown.



Identify if the given information is sufficient to determine whether the triangles are congruent. If so, state the congruence theorem or postulate.

	Given Information	Are the Triangles Congruent?	Theorem or Postulate Used
7.	$\overline{XG} \cong \overline{RB}, \angle RBL \cong \angle XGM, \overline{BL} \cong \overline{GM}$		
8.	$\angle BLR \cong \angle XMG, \overline{RL} \cong \overline{XM}, \angle BRL \cong \angle GXM$		
9.	$\overline{BL} \cong \overline{GM}, \angle BLR \cong \angle XMG, \overline{XG} \cong \overline{RB}$		
10.	$\overline{XG} \cong \overline{RB}, \overline{BL} \cong \overline{GM}, \overline{RL} \cong \overline{XM}$		

Triangle TWM is shown where $m\angle M = 35^\circ$, $m\angle T = 50^\circ$, and $WT = 5.2$ units.

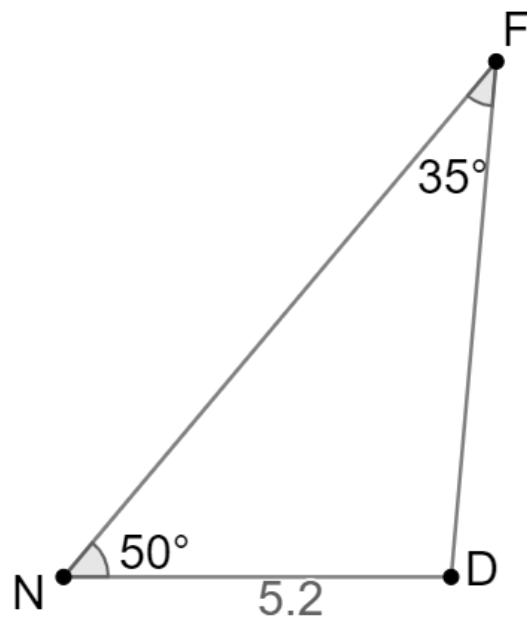


Complete the steps to describe how to construct a triangle that is congruent to $\triangle TWM$ using the angle-angle-side criteria where the first step is to draw a line that is longer than all of the sides of $\triangle TWM$.

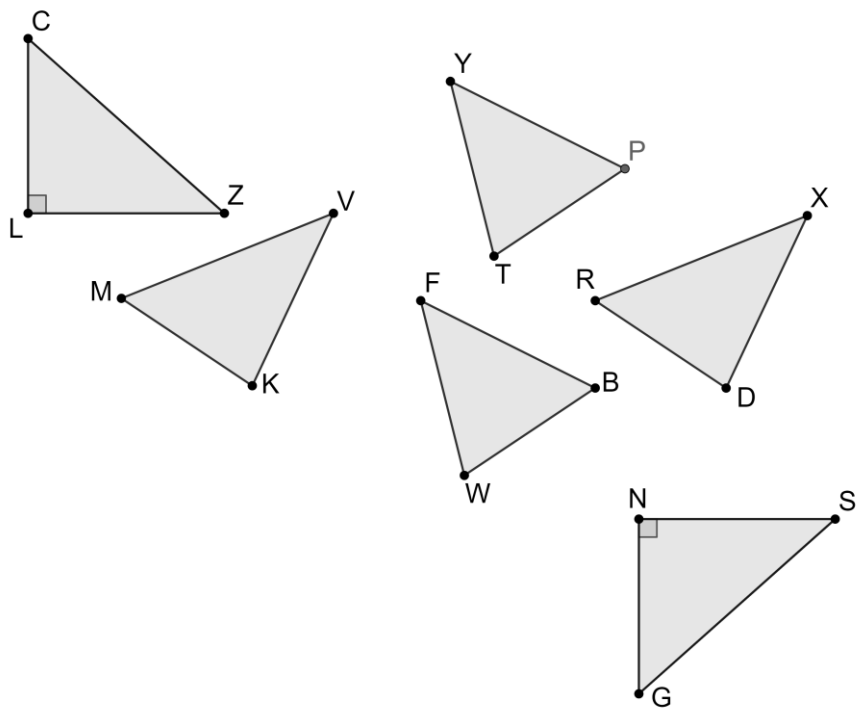
1. Use a compass or straightedge to copy _____. Label the endpoints as N and D .
2. Use a compass to copy angle _____. Construct a congruent angle using point N as the vertex.
3. Use a compass to copy angle _____. Construct a congruent angle so that one side of the angle goes through point D .

Shanice used the steps to create a triangle that is congruent to $\triangle TWM$. Her triangle is shown.

4. Describe what geometric tools Shanice can use to ensure $\triangle TWM \cong \triangle NDF$.



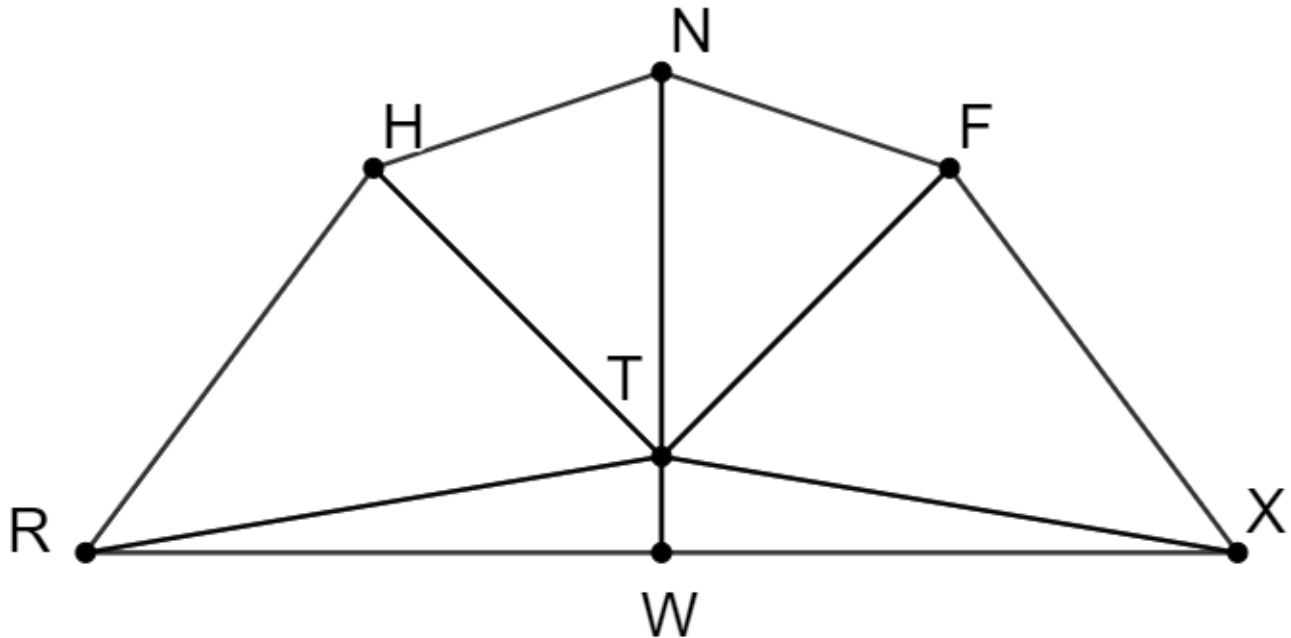
Triangles CLZ , MKV , YPT , BFW , GSN , and XDR are shown.



Determine if the given information is sufficient to determine whether the triangles are congruent. If so, state the congruence theorem or postulate.

	Given Information	Are the Triangles Congruent?	Theorem or Postulate Used
5.	$\overline{LZ} \cong \overline{NS}$, $\overline{CZ} \cong \overline{SG}$, $\angle LZC \cong \angle NSG$		
6.	$\angle PTY \cong \angle BWF$, $\overline{YP} \cong \overline{FB}$, $\angle TYP \cong \angle WFB$		
7.	$\overline{VK} \cong \overline{XD}$, $\overline{MK} \cong \overline{DR}$, $\angle KVM \cong \angle DXR$		
8.	$\overline{CZ} \cong \overline{GS}$, $\overline{CL} \cong \overline{GN}$		
9.	$\overline{MV} \cong \overline{XR}$, $\overline{KM} \cong \overline{RD}$, $\overline{XD} \cong \overline{YK}$		
10.	$\angle VMK \cong \angle XRD$, $\overline{VK} \cong \overline{XD}$, $\overline{MK} \cong \overline{DR}$		

Polygon $RHNFX$ is shown.



1. List all of the triangles in the diagram.

If $\triangle RTH \cong \triangle XTF$, complete the congruence statements.

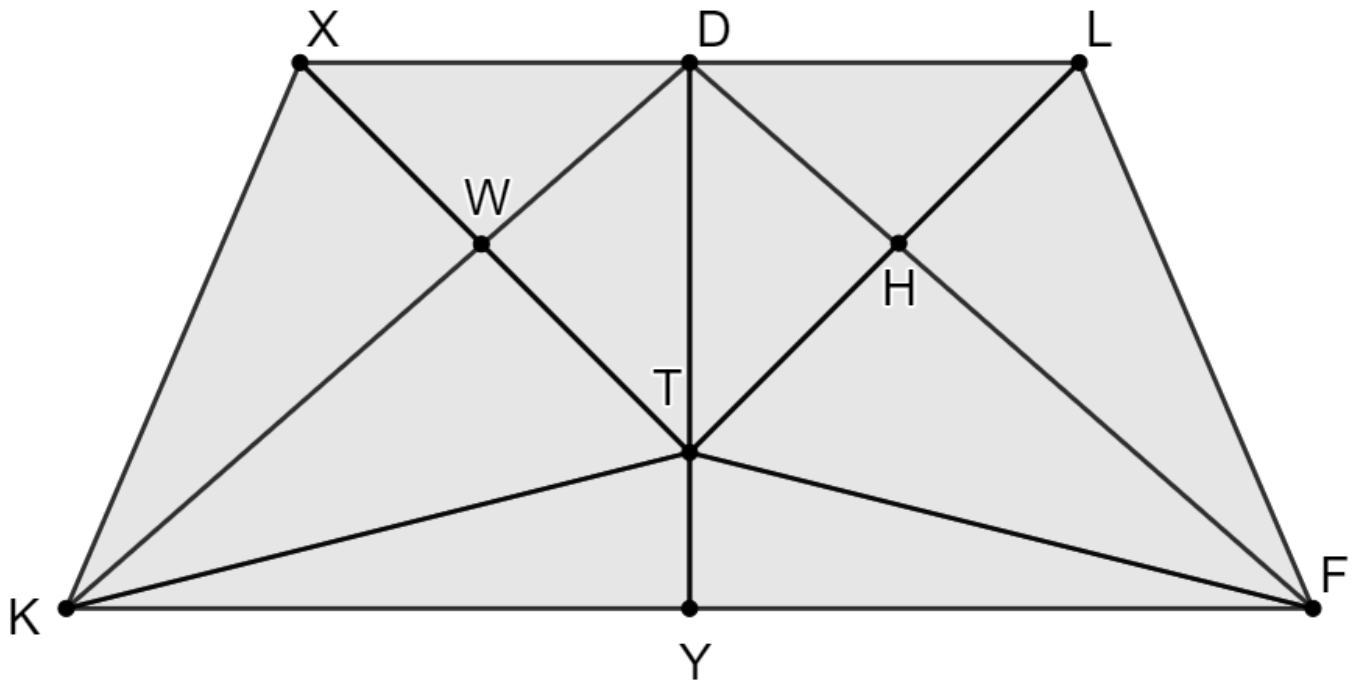
2. $\overline{TH} \cong$

3. $\angle TRH \cong$

If $\triangle RTW \cong \triangle XTW$, complete the congruence statement.

4. $\angle TXW \cong$

Quadrilateral $XLFK$ is shown.



If $\triangle XTD \cong \triangle LTD$, complete the congruence statements.

5. $\overline{XT} \cong$

6. $\angle DLT \cong$

7. $\overline{DT} \cong$

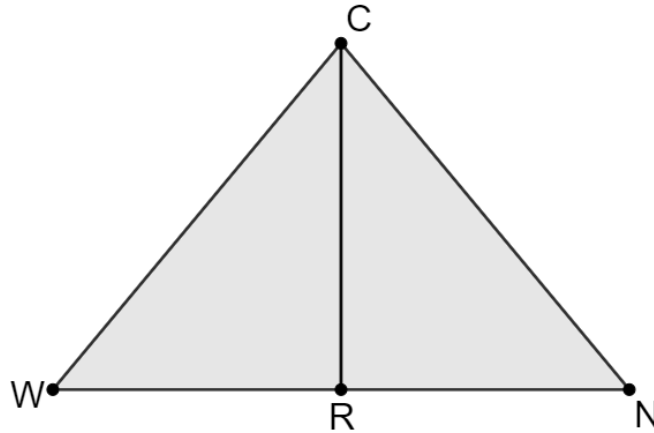
If $\triangle KDX \cong \triangle FDL$, complete the congruence statements.

8. $\overline{DF} \cong$

9. $\angle XDK \cong$

10. $\angle FLD \cong$

Triangle CWN is shown where R is the midpoint of $\overline{WN}$.



1. Triangle $\triangle CWN$ is an isosceles triangle.
Mark the diagram using the given information.
2. Which side(s) of triangles CRW and NRC are congruent because of the given information?
3. Which angle(s) of triangles CRW and NRC are congruent because of the given information?
4. Which side(s) or angle(s) of triangles CRW and NRC are congruent because of the reflexive property?
5. Using the given information, which if any of the triangle congruence theorems or postulates could be used to determine that the triangles are congruent.

☐ SSS

☐ SAS

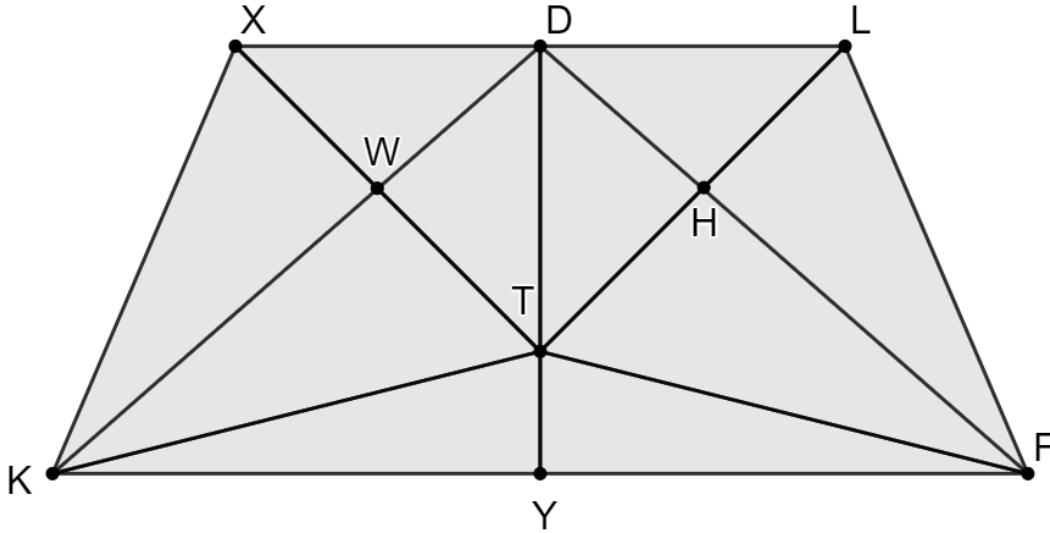
☐ ASA

☐ AAS

☐ HL

☐ none

Quadrilateral $KXLF$ is shown, where $\angle DKT \cong \angle TFH$, $\overline{KD} \cong \overline{DF}$, $\angle DTK \cong \angle FTD$, $\overline{XK} \cong \overline{LF}$, and $\overline{DY}$ is perpendicular to $\overline{KF}$.



6. Mark the diagram with the given information.

7. Complete the statement.

$\triangle \underline{\hspace{1cm}} \cong \triangle \underline{\hspace{1cm}}$ because $\overline{KD} \cong \overline{DF}$, $\angle DTK \cong \angle FTD$, and $\underline{\hspace{1cm}} \cong \underline{\hspace{1cm}}$.

8. Which postulate(s) or theorem(s) can be used to show that $\triangle KDY \cong \triangle FDY$?

☐ SSS ☐ SAS ☐ ASA ☐ AAS ☐ HL ☐ none

9. Which postulate(s) or theorem(s) can be used to show that $\triangle DKX \cong \triangle DFL$?

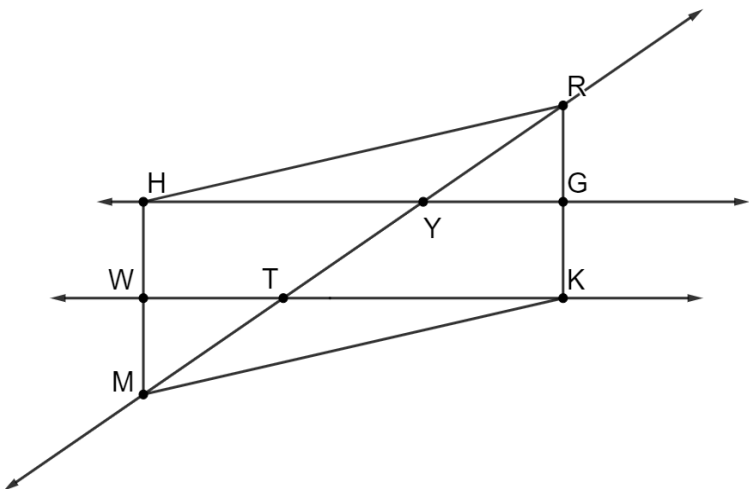
☐ SSS ☐ SAS ☐ ASA ☐ AAS ☐ HL ☐ none

10. Which postulate(s) or theorem(s) can be used to show that $\triangle TDF \cong \triangle TDK$?

☐ SSS ☐ SAS ☐ ASA ☐ AAS ☐ HL ☐ none

A diagram is shown where $\overline{HG} \parallel \overline{WK}$,
 $\overline{HR} \parallel \overline{MK}$, and $\overline{HR} \cong \overline{MK}$.

Use the given information to answer
 the questions about the incomplete
 proof that is shown.

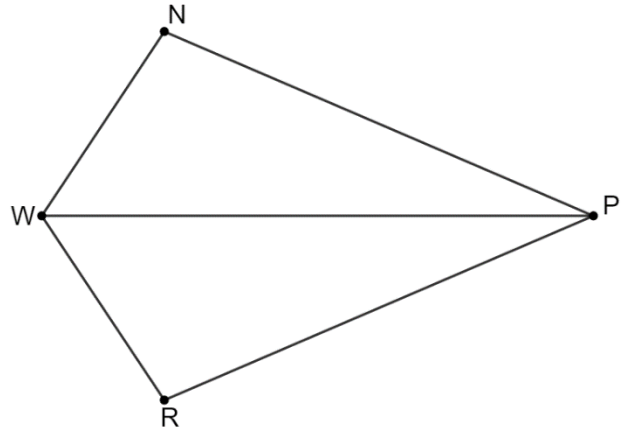


Statement	Reason
1. $\overline{HG} \parallel \overline{WK}$	1. Given
2.	2.
3. $\overline{HR} \parallel \overline{MK}$	3. Given
4.	4.
5. $\overline{HR} \cong \overline{MK}$	5. Given
6. $\triangle HYR \cong \triangle KTM$	6.

- What is the statement for line 2 of the proof?
- What is the reason for line 2 of the proof?
- What is the statement for line 4 of the proof?
- What is the reason for line 4 of the proof?
- What is the reason for line 6 of the proof?

A quadrilateral is shown where $\overline{WP}$ bisects $\angle NPR$, and $\overline{NP} \cong \overline{PR}$.

Use the given information to answer the questions about the incomplete proof that is shown.



Statement	Reason
1. $\overline{NP} \cong \overline{PR}$	1. Given
2. $\overline{WP}$ bisects $\angle NPR$	2. Given
3.	3.
4.	4. Reflexive property
5. $\triangle PRW \cong \triangle \underline{\hspace{1cm}}$	5.

6. What is the statement for line 3 of the proof?

7. What is the reason for line 3 of the proof?

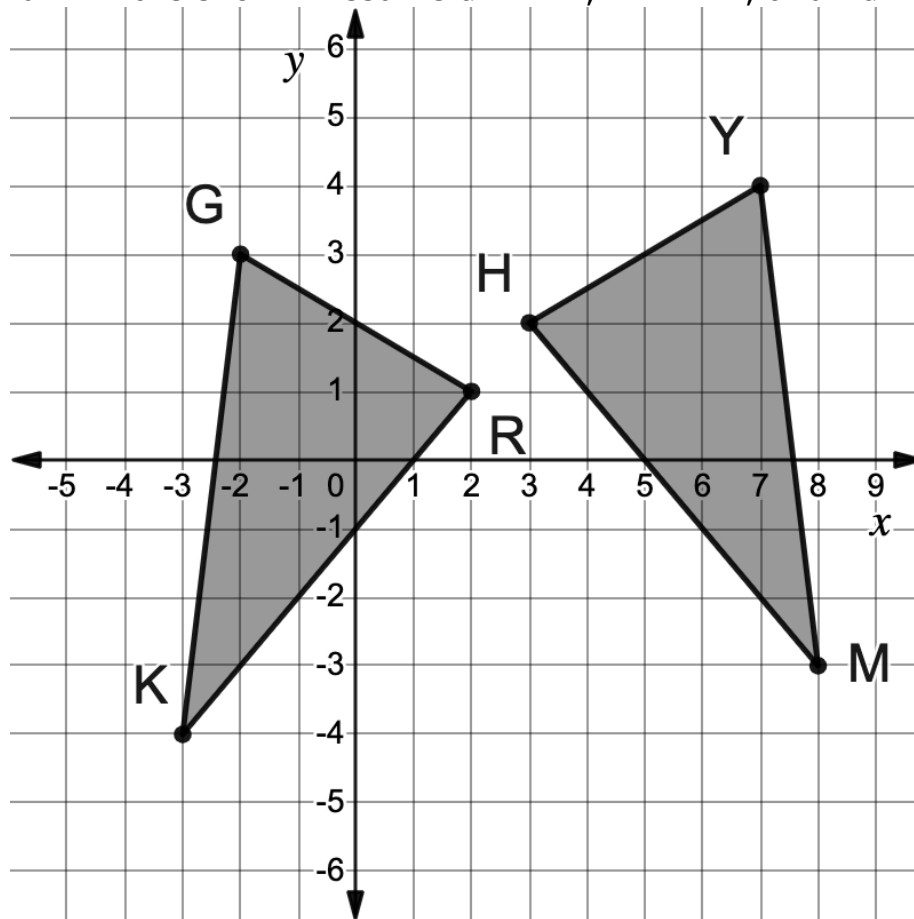
8. What is the statement for line 4 of the proof?

9. Complete the congruence statement for the statement of line 5 of the proof.

$$\triangle PRW \cong \triangle \underline{\hspace{1cm}}$$

10. What is the reason for line 5 of the proof?

Triangles GRK and MHY are shown. Assume $\overline{GR} \cong \overline{HY}$, $\overline{RK} \cong \overline{HM}$, and $\overline{KG} \cong \overline{YM}$.



1. Explain how to use geometric tools to verify that $\triangle HYM \cong \triangle RGK$.
2. Write a sequence of transformations that will map $\triangle HYM$ onto $\triangle RGK$.

Show your work for questions 3 through 5 on the coordinate grid.

3. Draw $\triangle H'Y'M'$ the result of the first transformation.
4. Map the vertices of $\triangle HYM$ to $\triangle H'Y'M'$.
5. Map the vertices of $\triangle H'Y'M'$ to $\triangle RGK$.

For questions 6 through 8, complete the statements.

6. The outcome of mapping the transformations of $\triangle HYM$ to $\triangle H'Y'M'$ and then from $\triangle H'Y'M'$ to $\triangle$ _____ shows that point H' coincides with point _____, point Y' coincides with point _____, and point M' coincides with point _____.

7. Because the definition of _____ in terms of _____
 _____ preserves _____,
 it can be stated that $\triangle HYM \cong \triangle$ _____.

8. Since it was assumed that $\overline{GR} \cong \overline{HY}$, $\overline{RK} \cong \overline{HM}$, and $\overline{KG} \cong \overline{YM}$ then _____ is sufficient to show the triangles are congruent.

9. Select all of the given statements that would result in a sequence of transformations that would show SAS congruence is sufficient to show the triangles are congruent.

- ☐ $\overline{GR} \cong \overline{HY}$, $\overline{RK} \cong \overline{HM}$, $\angle GRK \cong \angle YHM$
- ☐ $\overline{GR} \cong \overline{HY}$, $\overline{KG} \cong \overline{YM}$, $\angle GRK \cong \angle YHM$
- ☐ $\overline{KG} \cong \overline{YM}$, $\overline{RK} \cong \overline{HM}$, $\angle RKG \cong \angle YMH$
- ☐ $\overline{KG} \cong \overline{YM}$, $\overline{RK} \cong \overline{HM}$, $\angle RGK \cong \angle MYH$
- ☐ $\overline{RK} \cong \overline{HM}$, $\overline{GR} \cong \overline{HY}$, $\angle RGK \cong \angle MYH$
- ☐ $\overline{RK} \cong \overline{HM}$, $\overline{KG} \cong \overline{YM}$, $\angle GRK \cong \angle YHM$

10. Select all of the given statements that would result in a sequence of transformations that would show ASA congruence is sufficient to show the triangles are congruent.

- ☐ $\angle RGK \cong \angle MYH$, $\angle GRK \cong \angle YHM$, $\overline{RK} \cong \overline{HM}$
- ☐ $\angle RGK \cong \angle MYH$, $\angle GRK \cong \angle YHM$, $\overline{GR} \cong \overline{HY}$
- ☐ $\angle GRK \cong \angle YHM$, $\angle RGK \cong \angle MYH$, $\overline{KG} \cong \overline{YM}$
- ☐ $\angle RKG \cong \angle YMH$, $\angle RGK \cong \angle MYH$, $\overline{KG} \cong \overline{YM}$
- ☐ $\angle RKG \cong \angle YMH$, $\angle GRK \cong \angle YHM$, $\overline{RK} \cong \overline{HM}$
- ☐ $\angle RKG \cong \angle YMH$, $\angle RGK \cong \angle MYH$, $\overline{GR} \cong \overline{HY}$

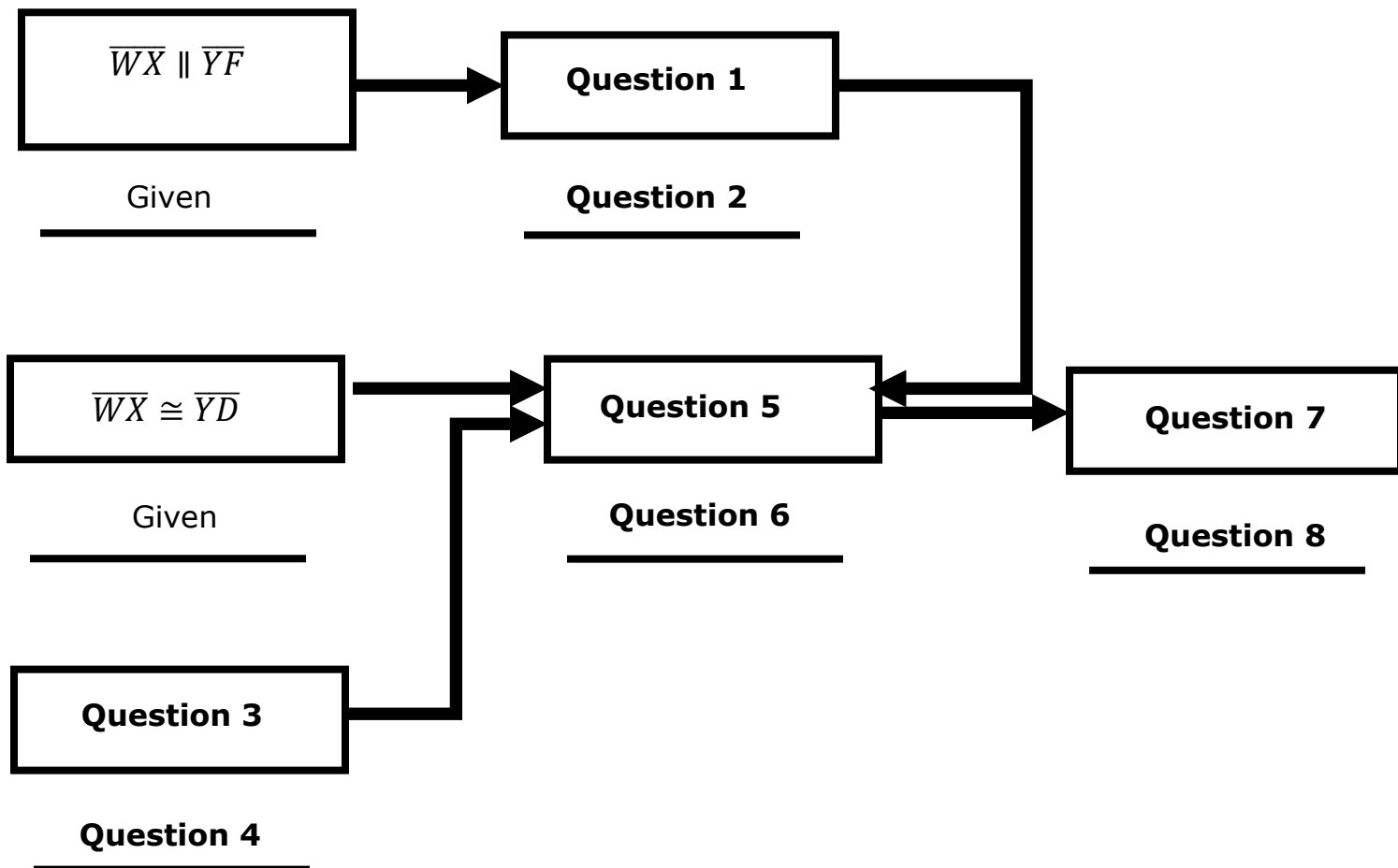
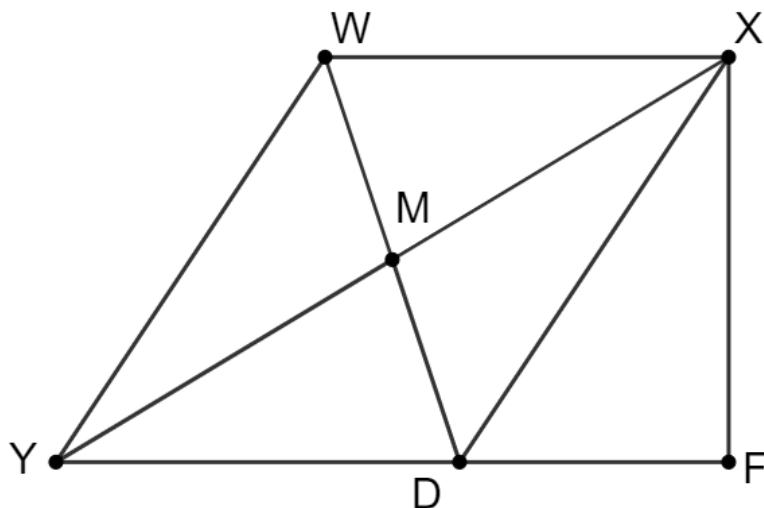
Geometry
Unit 5
Lesson 7 Practice

Name _____
Date _____
Period _____

Quadrilateral $WXYF$ is shown where
 $\overline{WX} \cong \overline{YD}$ and $\overline{WX} \parallel \overline{YF}$.

Given: $\overline{WX} \parallel \overline{YF}$, $\overline{WX} \cong \overline{YD}$
Prove: $\overline{WY} \cong \overline{XD}$

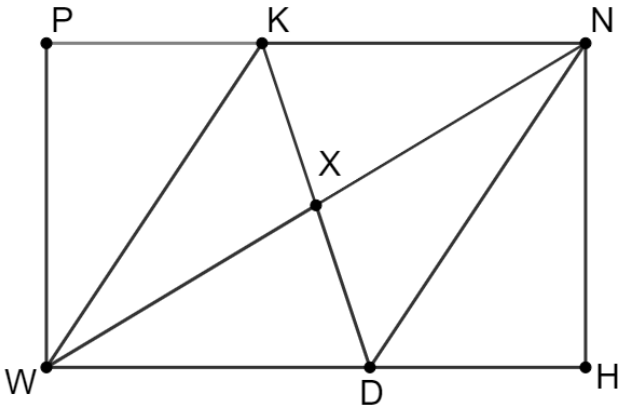
Use the given information to answer
the questions about the incomplete
proof that is shown. Each empty box
has the corresponding question
number.



1. What is the statement that belongs in the box?
2. What is the reason for the statement?
3. What is the statement that belongs in the box?
4. What is the reason for the statement?
5. What is the statement that belongs in the box?
6. What is the reason for the statement?
7. What is the statement that belongs in the box?
8. What is the reason for the statement?
9. Explain if it is possible to prove that $\overline{WM} \cong \overline{MD}$ using the same given statements.
10. Explain if it is possible to prove $\triangle WMX \cong \triangle DMY$ using the same given statements.

Quadrilateral $PNHW$ is shown where $\overline{KW} \parallel \overline{ND}$ and X is the midpoint of $\overline{WN}$

Given: $\overline{KW} \parallel \overline{ND}$, X is the midpoint of $\overline{WN}$
 Prove: $\overline{WK} \cong \overline{DN}$



Use the given information to answer the questions about the incomplete proof that is shown.

Statement	Reason
1. $\overline{KW} \parallel \overline{ND}$	1. Given
2.	2.
3. X is the midpoint of $\overline{WN}$	3. Given
4.	4.
5.	5.
6.	6.
7. $\overline{WK} \cong \overline{DN}$	7.

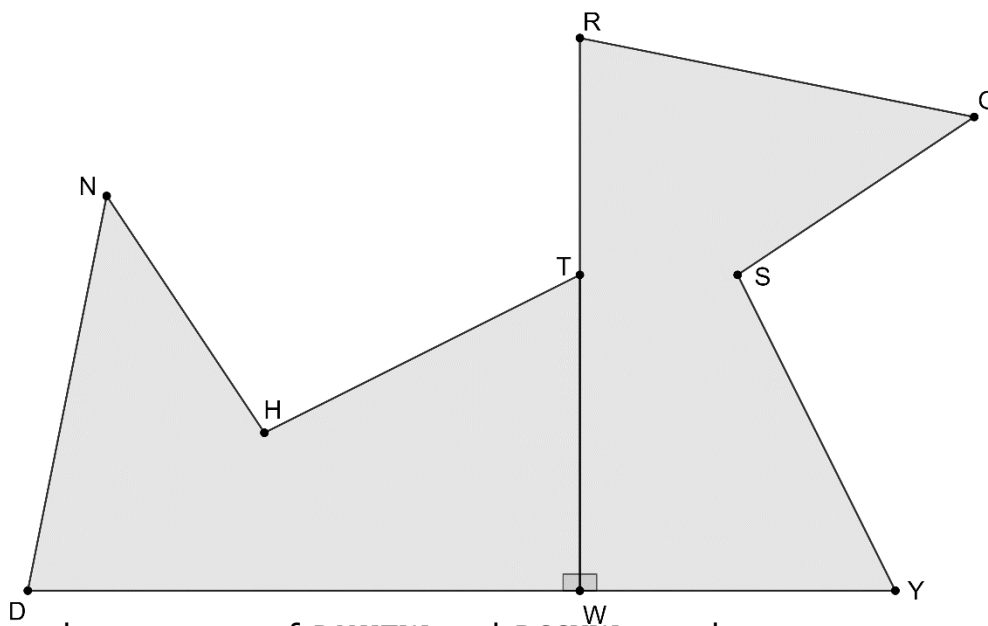
- What is the statement for line 2 of the proof?
- What is the reason for line 2 of the proof?

3. What is the statement for line 4 of the proof?
4. What is the reason for line 4 of the proof?
5. What is the statement for line 5 of the proof?
6. What is the reason for line 5 of the proof?
7. What is the statement for line 6 of the proof?
8. What is the reason for line 6 of the proof?
9. What is the reason for line 7 of the proof?
10. Explain if it is possible to prove that $\triangle WXD \cong \triangle NXK$ using the same given statements.

Geometry
Unit 5
Lesson 9 Practice

Name _____
Date _____
Period _____

Congruent polygons $DNHTW$ and $RGSYW$ are shown, where $\angle DWT$ and $\angle RWY$ are right angles.



Some of the angle measures of $DNHTW$ and $RGSYW$ are shown.

$$m\angle HTW = (6x - 15)^\circ$$

$$m\angle NHT = (12k + 14)^\circ$$

$$m\angle DNH = (3x + 6)^\circ$$

$$m\angle WRG = (3k + 13)^\circ$$

$$m\angle WYS = (4x + 11)^\circ$$

$$m\angle YWR = (4k + 6)^\circ$$

1. Determine the value of k .

2. Determine the value of x .

Determine each angle measure.

3. $m\angle GSY = \underline{\hspace{2cm}}$

4. $m\angle RGS = \underline{\hspace{2cm}}$

5. $m\angle WDN = \underline{\hspace{2cm}}$

Some of the measures of the side lengths of $DNHTW$ and $RGSYW$ are shown.

$$ND = 5h + 33$$

$$HT = 7h - 33$$

$$RW = 9h - 6$$

$$GS = 2h + 12$$

$$TW = 5h$$

$$DW + WY = 330$$

6. Determine the value of h .

7. What is the length of $\overline{HN}$?

9. What is the length of $\overline{RG}$?

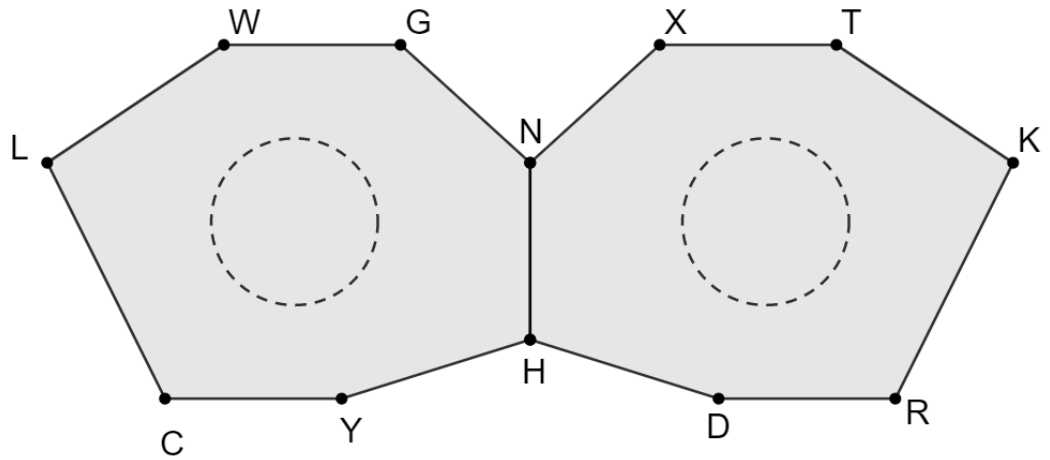
8. What is the length of $\overline{SY}$?

10. What is the length of $\overline{DY}$?

Geometry
Unit 5
Lesson 10 Practice

Name _____
Date _____
Period _____

A landscape architect is designing an area in a park that will have two fountains with benches for sitting around the fountains. The two areas are congruent polygons. The architect created a diagram where congruent polygons $LWGNHYC$ and $KTXNHDR$ represent the two areas that surround the fountains.



In polygon $KTXNHDR$, $\overline{NX} \cong \overline{NH} \cong \overline{DR} \cong \overline{XT}$. The perimeter of polygon $LWGNHYC$ is 564 inches (in.)

Some of the side measures of polygons $LWGNHYC$ and $KTXNHDR$ are shown.

$$HD = 20x - 14.4$$

$$GW = 13x + 9.6$$

$$CL = 18x + 21.6$$

$$KT = 9x + 43.2$$

- Write an equation that can be used to determine the value of x .
- Determine the value of x .
- Determine the length of side HN .
- Determine the length of side RK .
- Determine the length of side XN .

Some of the angle measures of polygons $LWGNHYC$ and $KTXNHDR$ are shown.

$$m\angle CLW = (2k + 55)^\circ$$

$$m\angle WGN = (7k - 9)^\circ$$

$$m\angle YHN = (5k + 2)^\circ$$

$$m\angle XTK = (9k - 43)^\circ$$

$$m\angle HYC = m\angle NXT + 25^\circ$$

$$m\angle KRD = m\angle TKR + 20^\circ$$

6. Given that $m\angle XTK + m\angle TKR + m\angle KRD = 360^\circ$, determine the value of k .

Determine the measure of each angle.

7. $m\angle NXT = \underline{\hspace{2cm}}$

8. $m\angle HYC = \underline{\hspace{2cm}}$

9. $m\angle DHN = \underline{\hspace{2cm}}$

10. $m\angle LWG = \underline{\hspace{2cm}}$